Ecole Supérieure d'Informatique et du Numérique
COLLEGE OF ENGINEERING & ARCHITECTURE

Lab 06: Fundamental Security **Instructor:** Prof. Anass Sebbar **Level:** Engineering - 3rd Year

Lab 06: Introduction to iptables & Firewalls

Objective

This lab proposes to install a firewall on a Linux machine in order to separate two different network segments and to implement the necessary security policy.
To do this, we will proceed according to the following steps:
- Configuration of the network parameters, remote and firewall of a Web server and a client machines;
- Use of Nmap
- Secure a web server by the configuration of the Iptables Firewall;

Instructions

- The lab report must be submitted one week after the lab session on the Moodle platform, respecting the deadline mentioned there.
- The lab must be performed individually in class, but the report should be submitted in groups of up to 2 students.
- Lab groups must remain the same for all reports throughout the semester

Introduction

Linux uses netfilter kernel module for firewall. The netfilter module in kernel allows us to filer any incoming, outgoing and forwarded data packet even before it reaches at user level application. To communicate with netfilter module we can use iptables.

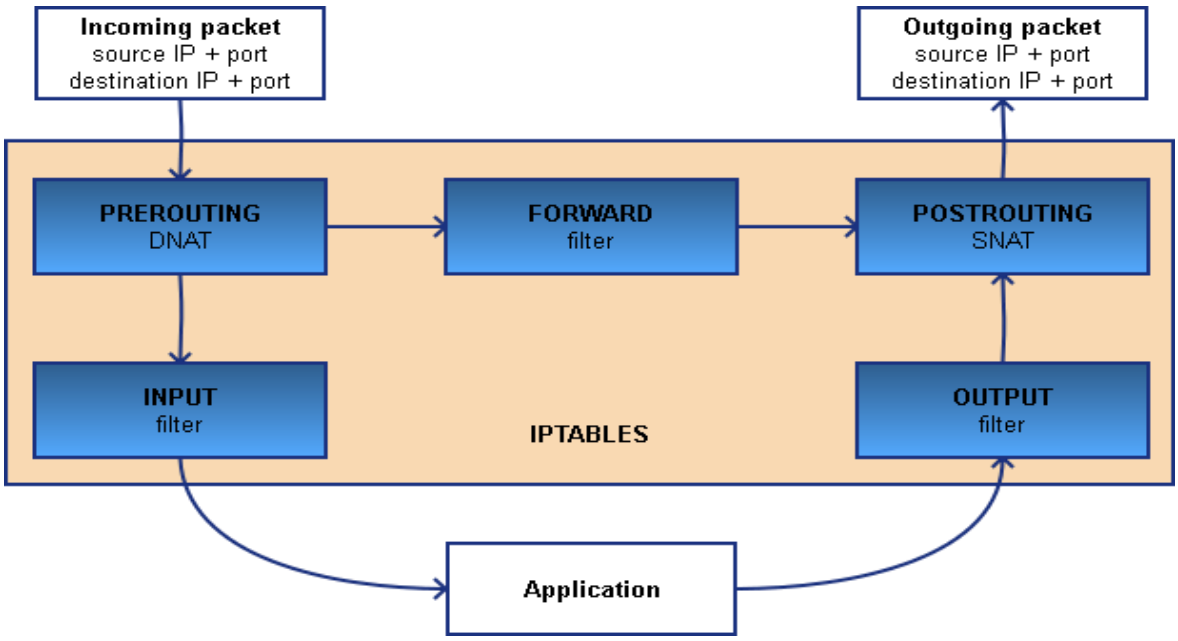

Figure 1: iptables architecture

As we know iptables filter data packets based on pre-defined rules. So basically rules are the instructions which iptables should follow when a match found. Similar rules are grouped in chains. For examples we may group rules in two chains, in first chain we can keep the rules which will filter the incoming traffic while in second chain we can store the rules which will filter the outgoing traffic. There are five pre-defined chains; INPUT, OUTPUT, FORWARD, PREROUTING and POSTROUTING. If iptables are configured, data packet must go through at least one of these chains. Each chain has an action known as target. Target have following values

- ACCEPT :- Allows the packet pass through.
- REJECT :- Discards the packet and sends a notification back.
- DROP :- Discards the packet without sending notification back.
- LOG :- Sends packet information to rsyslogd daemon for logging.

Chain can also inspect the packet based on its state. Following are some common states:-

- NEW :- Packet is not the part of any exiting communication.
- ESTABLISHED :- Packet is the part of an existing communication.
- RELATED:- Packet is the part of other existing communication.
- INVALID :- Packet does not match with any states.

1 Securing a web server with iptables

1.1 Setting up the client and the server

1.1.1 Set up two machines in the same network :

In your virtual environment, set up your machine to be on the same network (either both NAT or Bridged or Host only) .

1.1.2 What is the ip address of your two machines ?

1.1.3 Use the ping command to check if the server is reachable from the client.

1.1.2 Start an http server in the server machine

1- Start a web server to serve the page index.html
Sudo service appache2 start

check if the server is serving the index page by opening the following URL in your browser:

- Client: http://<ip of the server>:80
- Server: http://127.0.0.1:80

1.1.3 Install the required tools

In this lab we need three tools: Nmap in the client machine and and ssh server in ther server

Command Line

 # In the client machine install nmap

```
user@client$  sudo apt−get update
user@client$  sudo apt−get install –y nmap
```

#In the Server machine install SSH

```
user@server$ sudo  apt−get update
 user@server$ sudo  apt-get install -y openssh-server
```

- Start ssh server with sudo service start ssh
- From the client machine check if you can access the server with ssh

$ ssh (user name)@<server ip address>

1.2 Listing the open ports in the server

1- From the client side , open a new terminal and scan ports with nmap :

$ nmap <the ip of your server >

1. From the server side

$ sudo netstat −tulpn

1.3 Secure the webserver

We will create a configuration file that will be used to configure our firewall in the server machine. Create a new file config_firewall.sh using you file editor :

```
#! /bin/bash

echo -n "clearing chaines ... "
# deleting all rules
iptables -F INPUT
iptables -F OUTPUT
echo "done"
echo -n "defining default
policies ... "
iptables -P OUTPUT ACCEPT
iptables -P INPUT DROP
echo "done"
```

We need to execute the file as a superuser by executing the following commands:

```
user@server$    chmod +x config_firewall.sh
user@server$    sudo ./config_firewall.sh
```

Check the rules of the firewall using the following command:

iptables -L

Question2 :

2-1 Ping your server from the client machine.

2-2 Run an nmap scan from the client machine.

2-3 Browse the web server from the client machine.

Is the server still reachable in each of these cases.

All the traffic is blocked to the server. In the following, we will insert each rule at the end of the script config_firewall.sh and re-execute it. Use ping, ssh, nmap and browse to the server URL to check the effect to rule.

- **Allow traffic on loopback and established connections**

```
iptables -A INPUT -i lo -j ACCEPT
iptables -A INPUT -m state --state RELATED,ESTABLISHED -j ACCEPT
```

- **Allow ping requests**

```
iptables -A INPUT -p icmp --icmp-type echo-request -j ACCEPT
```

- **Allow traffic to the webserver**

```
iptables -A INPUT -p tcp --dport 80 -j ACCEPT
```

- **Allow traffic to the SSH server**

```
iptables -A INPUT -p tcp --dport 22 -s <ip_client>  -j ACCEPT
```